



Networking Fundamentals 4CC523



Week 4 – Network Design



Individual Health & Safety Responsibilities

To help prevent the spread of the COVID-19 virus we ask you to carry out the following, to help keep everyone safe & aid with contact tracing.

- If you have any COVID-19 symptoms leave the class immediately if possible & return to your accommodation. Contact the NHS on 111 & then contact student wellbeing.
- Wash your hands or sanitise them upon entering/exiting the area.
- Sign in/register via the Tap In system or alternative system in place.
- Identify a seating area & log the desk number on 'Who Sat Where?' on the Udo app. For people who don't have access to the app inform your lecturer to use the alternative system.
- Always maintain social distancing.
- When the class finishes, please use the cleaning materials provided to clean down the area you have used & then wash or sanitise your hands.
- We welcome the wearing of face coverings in teaching areas where possible.

Thanks for your help and continued support.

IMPORTANT INFO!!

–BE AWARE – BIG BROTHER IS WATCHING
AND LISTENING...

- Actually listening but not watching
- Inform me if you object to recordings being made
(and remind me every session!!)

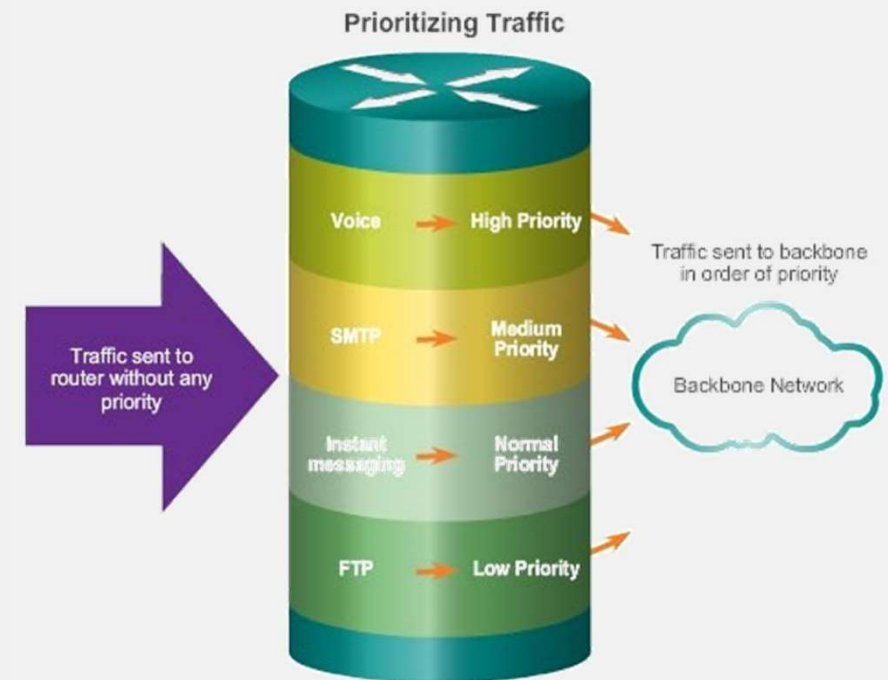
–TAP, TAP – only you and do stay for the lecture

Week 4: Objectives

- Design fundamentals
- Topologies
- Broadcast and Collision Domains
- Switching (L2)
- Security, Threats, Resilience and Reliability
- Redundancy
- Firewalls
- Hierarchical design
- Routing and Switching (L3)
- Address Allocation
- Expanding and extending the LAN

Design Considerations for a Small Network

- Requirements:
 - Internal Accessibility
 - External Access (out and in)
 - Logical Security
 - Physical Security
 - High transfer speed, Minimal delays
 - Reliable and Resilient
 - Adaptable to future needs

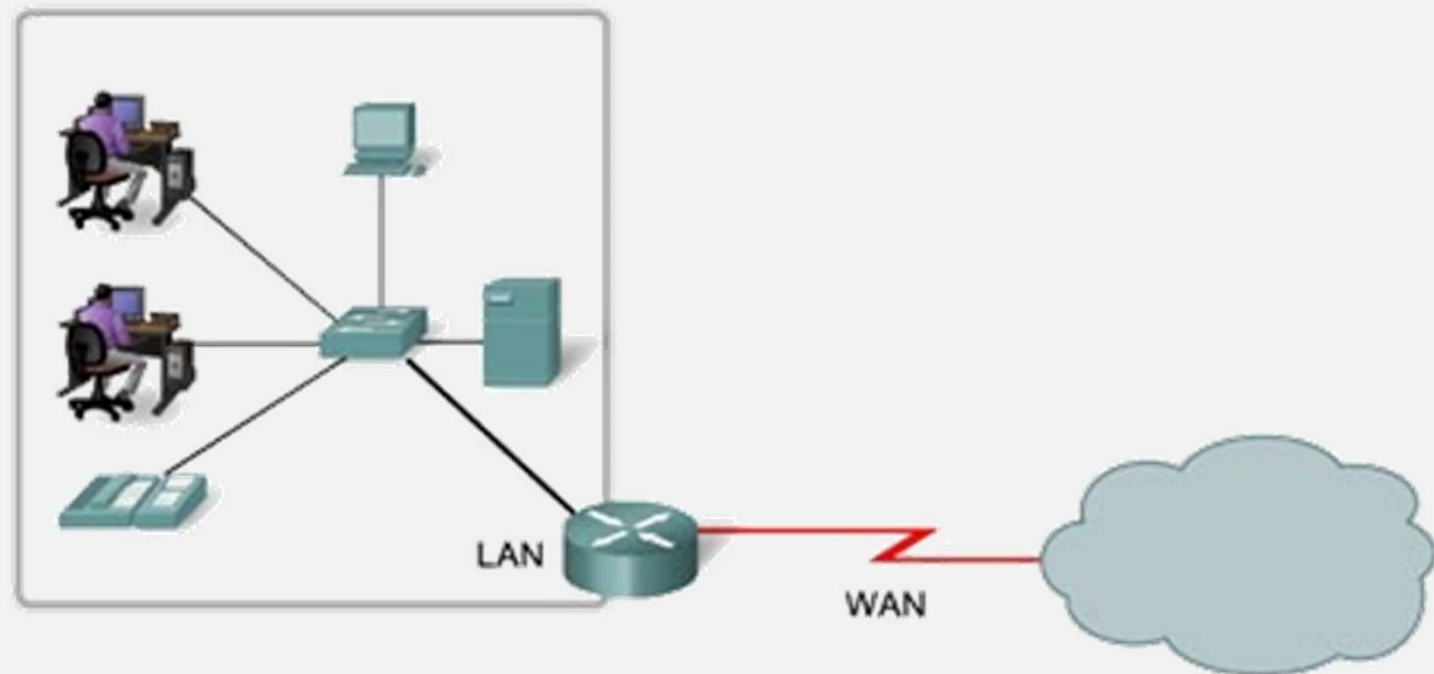


Design Fundamentals

- Network Topology (star, ring, bus...)
- Logical Topology (design concept)
- Physical Topology (how you build it)
- Addressing Scheme (how you access it)
- Routing / forwarding (advanced topic)

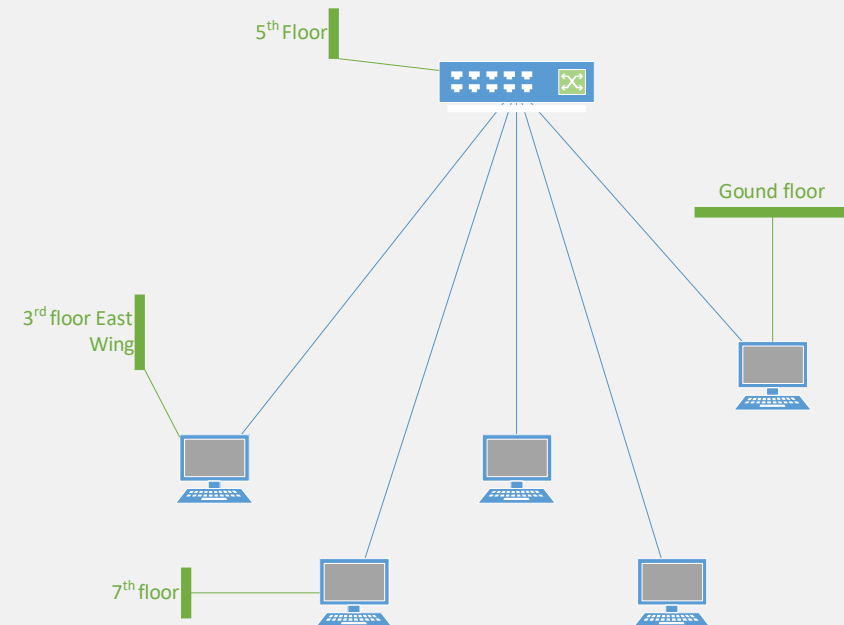
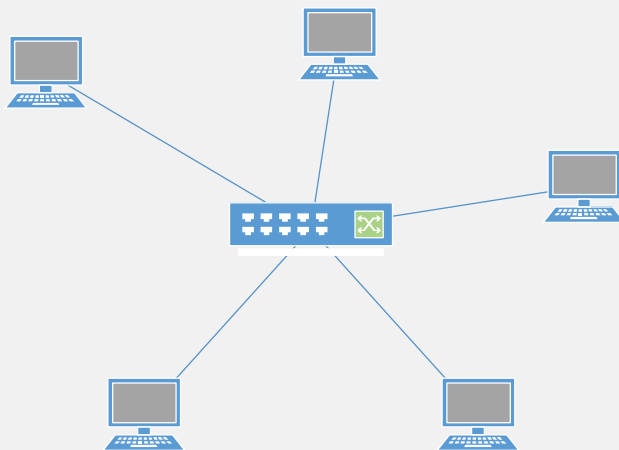
Small Network Topologies

- Typical Small Network Topology
- Star LAN with routed link to Internet (or “head office”)



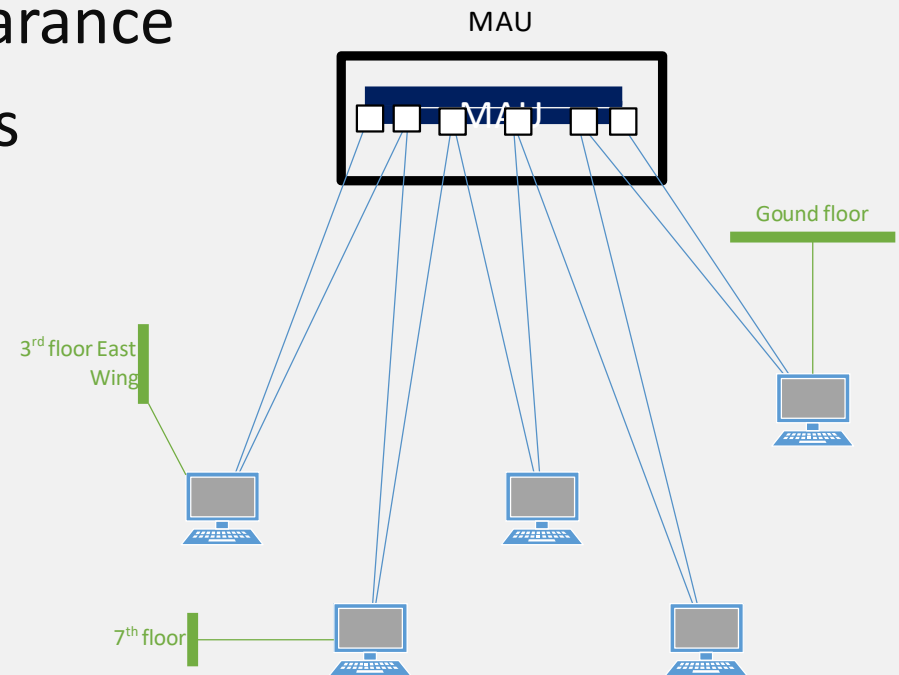
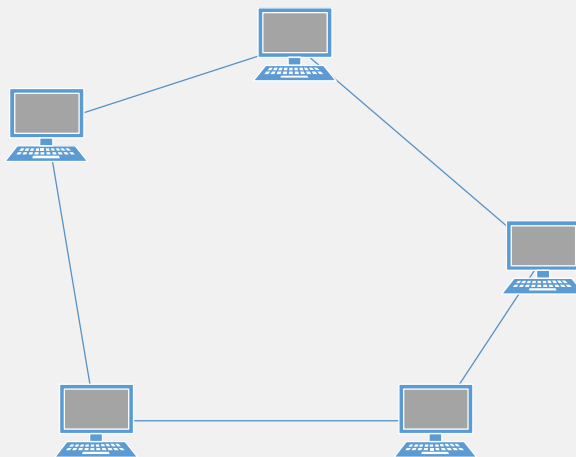
Basic LAN Topologies - Star

- (hub and spoke) – all devices connect to a central point
 - Very common for data networks
 - The “single logical point” could be a collection of physical devices
 - This is about physical connection BUT this is not the same as physical layout



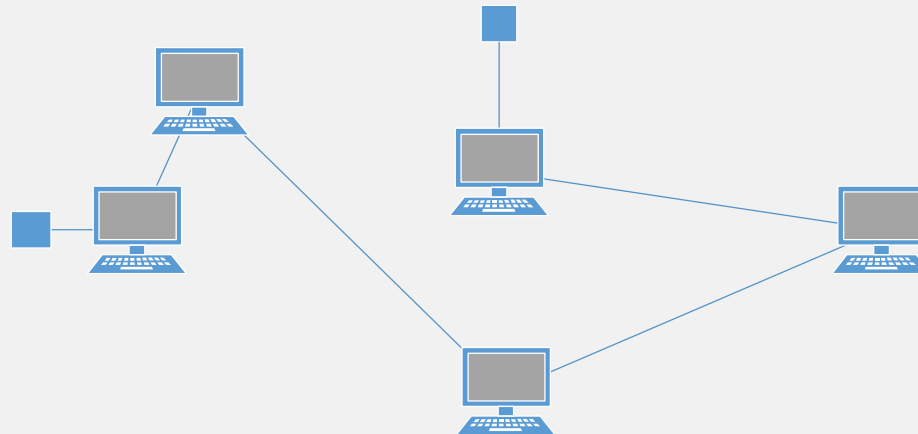
Basic LAN Topologies - Ring

- Devices are connected in a physical and logical ring
 - Again not the same as a physical layout
 - some ring networks look like Star connections, many have no identifiable “ring like” appearance
 - Extremely rare in data networks



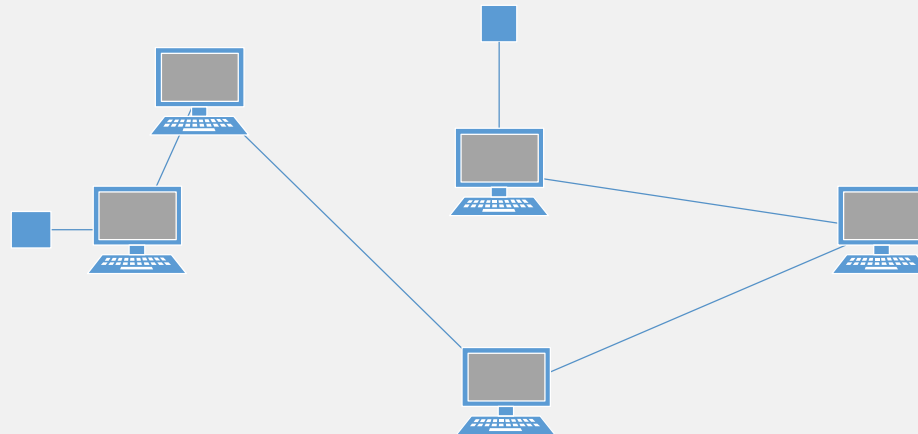
Basic LAN Topologies - Bus

- All devices attach to the same medium (wire) and share the access bandwidth
 - Only one message at a time!
 - Cheap implementation but very slow and potentially insecure
 - Used in control systems including Automotive applications
 - Physical connection but not layout



Basic LAN Topologies - Wireless

- All devices attach to the same medium (airwaves) and basic systems share the access bandwidth
 - Only one message at a time!
 - Cheap implementation but very slow and potentially insecure
 - Access speed depends on transmit power, distance, physical obstructions, other devices, local interference, ...



Collision and Broadcast Domains

- In most communication networks there is some “contention”
 - More than one device needs access
- In Ethernet based networks we need to consider two differentiating concepts
 - Will talk about how these are actually used another time

Collision Domains

- A collision domain is the LAN segment where devices must compete to communicate
- Only one message can be “on the wire” (or in the air) at any one time
- If more than one device attaches to the medium simultaneously, there will be a signal / message collision

Collision Domains

- Every device connected to a single wireless point channel (WR or WAP) is in a single collision domain
- Every device attached to a HUB (aka Repeater) is in a single collision domain
 - A big reason why we do not use them any longer
- Every port of a SWITCH is a collision domain **on its own**
- A **SWITCH breaks** the LAN segment into smaller **collision domains**, reducing device competition, increasing network response

Broadcast Messages

- Broadcast messages are important in LAN communication, sometimes a message is intended for every device, sometimes for an unknown device
 - When a device first attaches to the network it might not have an IP address. Its first action will be to “shout” for help. This will be a broadcast message to ask if any device can supply an address (is a dhcp server)
- A **broadcast domain** is the extent of the network where a broadcast frame can be “heard”
 - This is our cloud (don’t tell the other clouds what we are doing!)

Broadcast Domains

- Switches forward broadcast (and multicast) frames to all ports - switches **do not** break broadcast domains
- All ports of an unmanaged switch (or managed switches with default configuration) belong to the same broadcast domain
- If two or more switches are connected, broadcasts will be forward to all ports of all switches (except for ports that originally receive the broadcast)
- ROUTERS do not pass on broadcasts, they **break broadcast domains**
 - Amazingly clever!

Switching

- Switching is *traditionally* considered to be a Layer 2 function
- Operates in the LAN environment
- Operates only in a single Broadcast Domain
- Addressing is based on hardware / MAC address
- Forwards Ethernet Frames
- No routing
- No firewall
 - There are Layer 3 switches



Alleviating Network Congestion

- Switches reduce network congestion by:
 - Splitting a LAN into separate collision domains (no contention)
 - providing full-duplex communication between devices
- Switches have high port density and a low per-port cost
- Switches improve data throughput
 - Have high speed ports
 - Can buffer large frames (briefly hold data) to avoid collisions
 - Use fast hardware-based switching processes internally

Device Selection for a Small Network

- Factors to be considered when selecting intermediary devices



COST



PORTS



SPEED



EXPANDABLE/ MODULAR



MANAGEABLE

"Form Factor"

- Most commercial / Industrial equipment is intended to be mounted in a "19 inch network cabinet"
- Mounting is 19" wide and measured in "U" heights, rectangular stackable shapes minimise lost space



- Most consumer / small office / home office (soho) equipment is intended to sit on a desk or filing cabinet – irregular, not stackable
- Not very interchangeable!

"Form Factor"

Fixed Configuration Switches



Features and options are limited to those that originally come with the switch.

Stackable Configuration Switches



Stackable switches, connected by a special cable, effectively operate as one large switch.

Modular Configuration Switches



The chassis accepts line cards that contain the ports.

Physical / Operational Concerns

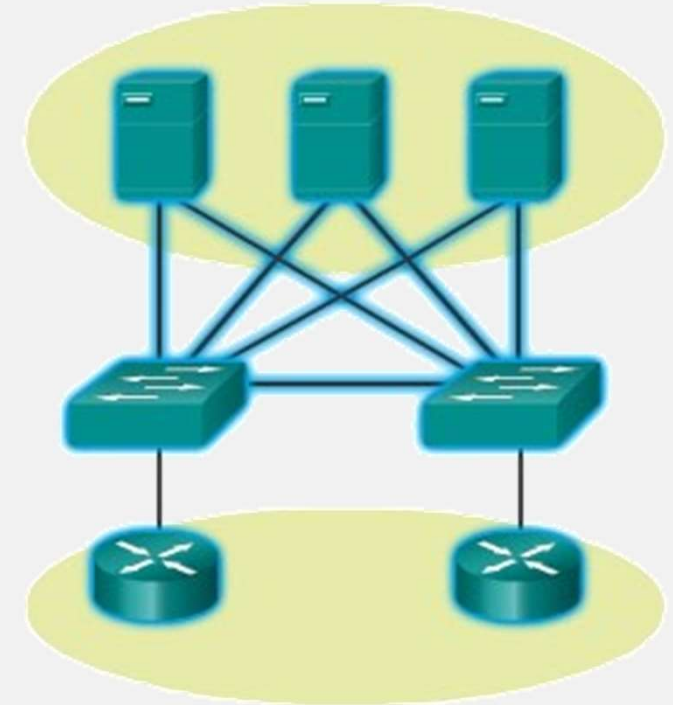
- Hardware threats
 - physical damage to servers, routers, switches, cabling and workstations. Theft.
- Environmental threats
 - temperature extremes (too hot or cold), humidity extremes (too wet or dry)
- Electrical threats
 - voltage spikes, unconditioned power (noise), insufficient supply voltage (brownouts), and total power loss
- Maintenance threats
 - poor handling (electrostatic discharge), lack of critical spare parts, poor cabling and poor labelling

Network Resilience / Reliability

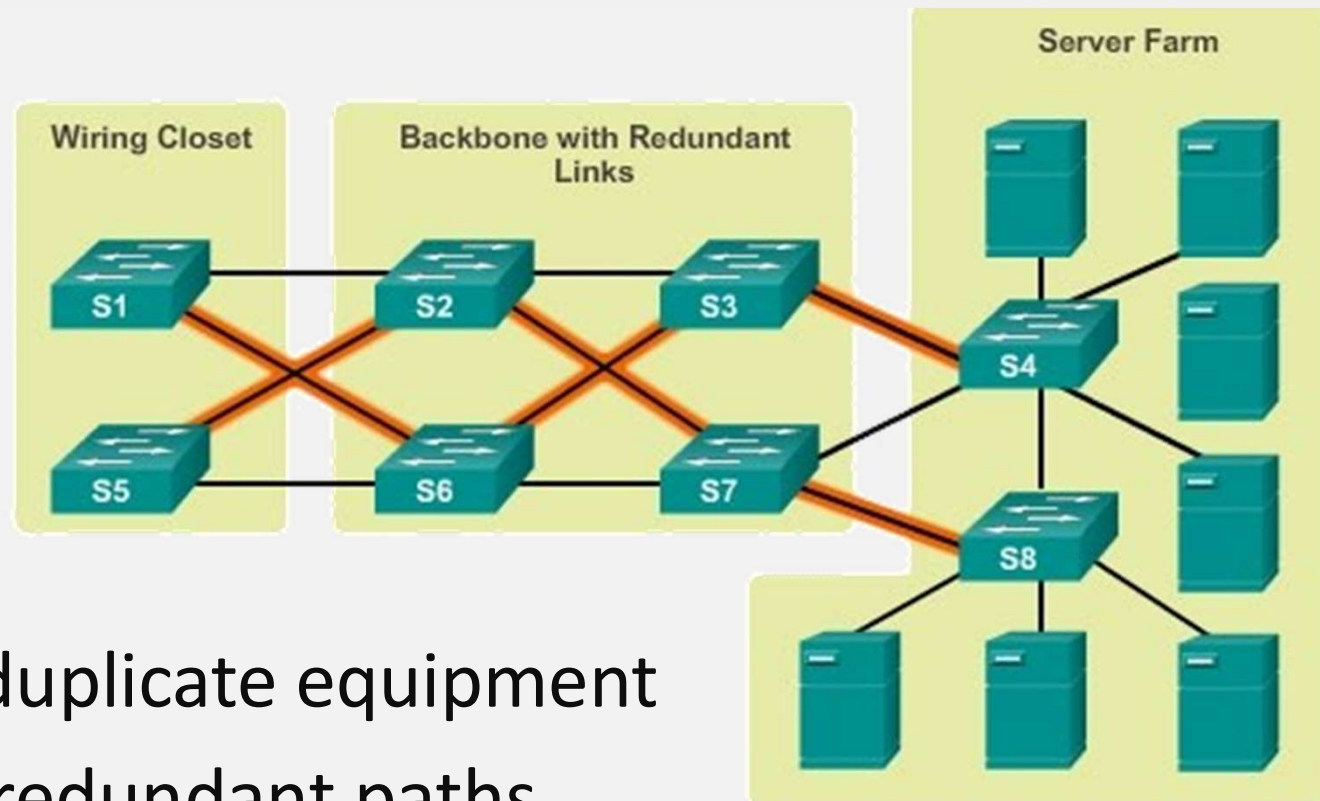
- Networks will be prone to failures if there are single points of failure
- Some single points are tolerable
 - single user access
- Some single points are not
 - Access to the user authentication server, mail server, possibly Internet

Redundancy in a Small Network

- Duplication of equipment and multiple links introduces “Redundancy”
 - Typically unused “network paths” that are automatically available if/when the main path fails
- Redundancy helps to eliminate single points of failure
- Improves the reliability of the network



Planning for Redundancy



- Installing duplicate equipment
- Providing redundant paths

Firewalls

- A firewall sits between two or more networks. It controls traffic and helps prevent unauthorised access from less trusted networks
- Packet Filtering
- Application Filtering
- URL Filtering
- Stateful Packet Inspection (SPI) - Incoming packets must be legitimate responses to requests from internal hosts



Cisco Security Appliances



Server-Based Firewall



Linksys Wireless Router with Integrated Firewall



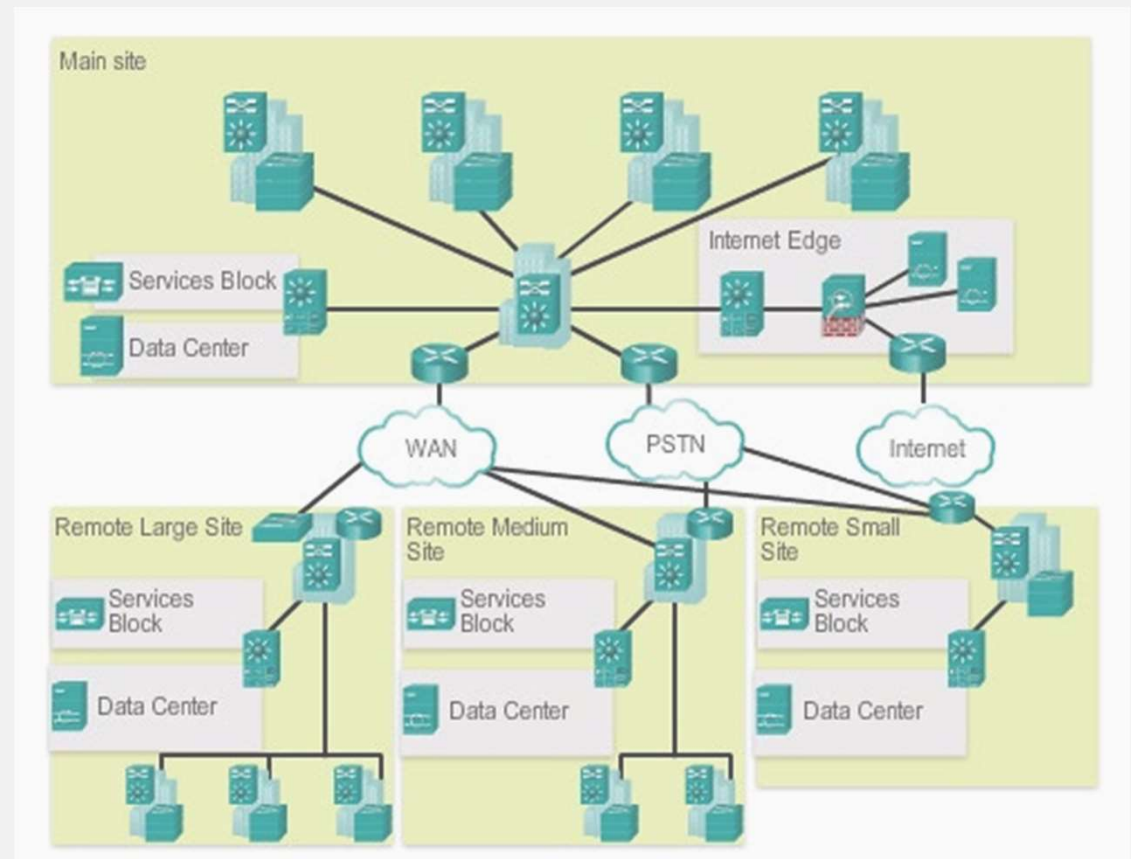
Personal Firewall

Firewalls

- The two networks may be our (trusted) network and the (totally untrusted) Internet
- The networks may also be two of our LANs, preventing unauthorised, unexpected or undesirable access (accidental or otherwise)
 - E.g. Personnel records, payroll department, records and fees, the exam papers answer repository, student machines
- Could also be the access network and the internal network of our PC

Network Scale

- Not absolute values but typically:
- Small network
~ 1 to 200 users
- Medium network
~ 200 to 1,000 users
- Large network
~ 1,000+ users



Scaling a Small Network

- Important points to consider as the network grows:
 - Documentation – physical and logical topology
 - Device inventory – list of devices that use or comprise the network
 - Budget – itemised IT budget, including fiscal year equipment purchasing budget
 - Traffic Analysis – protocols, applications, and services and their respective traffic requirements should be documented

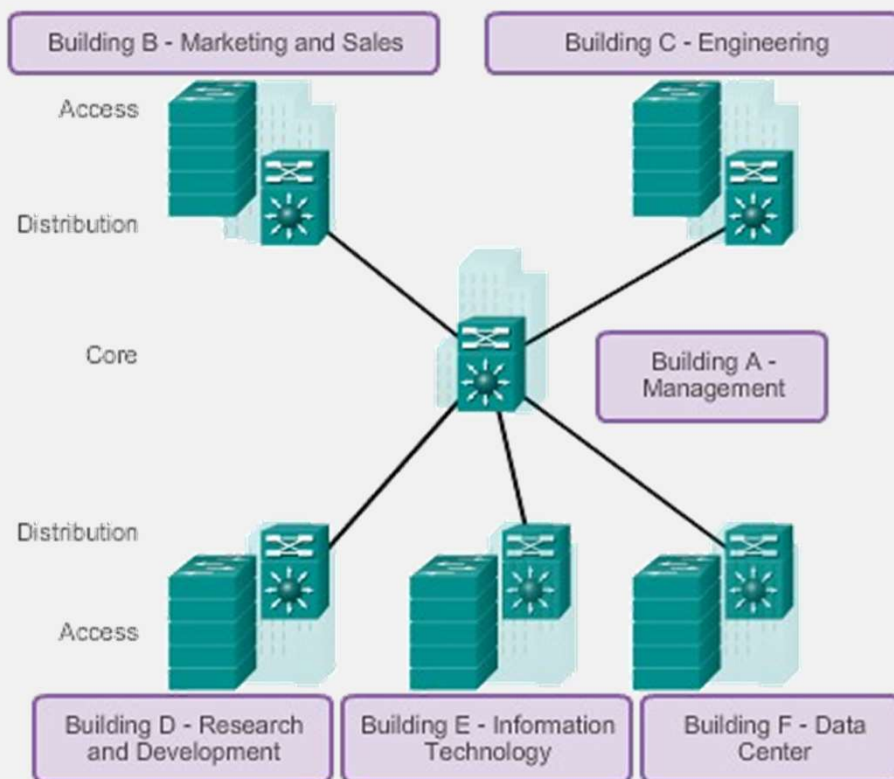
Addressing for a Small Network

- IP addressing scheme should be planned, documented and maintained based on the type of devices receiving the address.
- Examples of devices that will be part of the IP design:
 - End user devices
 - Servers and peripherals
 - Hosts that are accessible from the Internet
 - The intermediary devices
- Planned IP schemes help the administrator:
 - Track devices and troubleshoot
 - Control access to resources

Hierarchical Network Design

- 3 layer conceptual model for logical network design
- Access layer
- Distribution layer
- Core layer
- Use of layers varies depending on scale

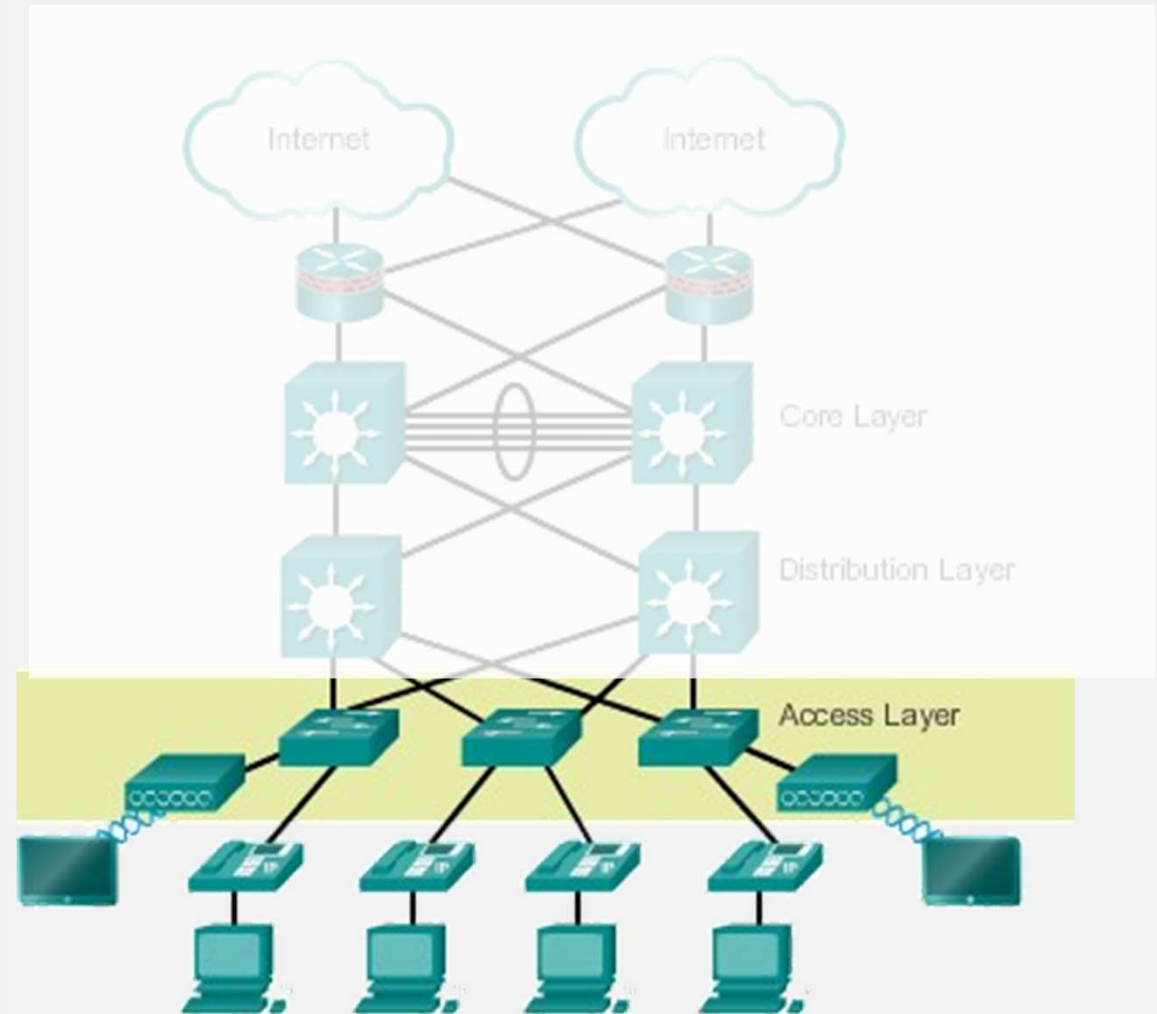
Hierarchical Network Design



- **Access layer** – Provides workgroup or user access to the network
- **Distribution layer** – Provides policy-based connectivity
- **Core layer** – Provides fast transport between distribution switches

Access Layer

- Layer 2 switching
- High availability
- Port security
- QoS classification and marking
- Power over Ethernet (PoE)



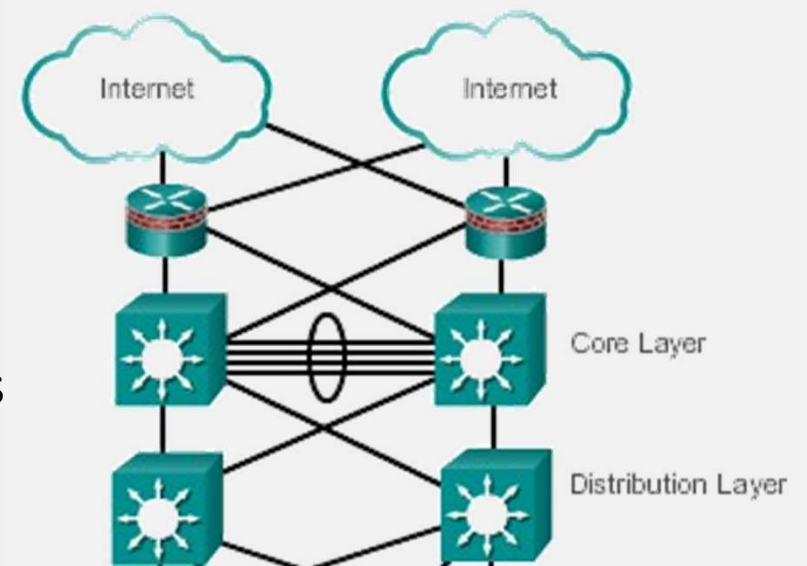
Distribution and Core Layers

- **Core layer**

- Provides high-speed switching (i.e., fast transport)
- Provides reliability and fault tolerance
- Scales by using faster (not more) equipment
- Avoids CPU-intensive packet manipulation caused by security, inspection, quality of service (QoS) classification, or other processes

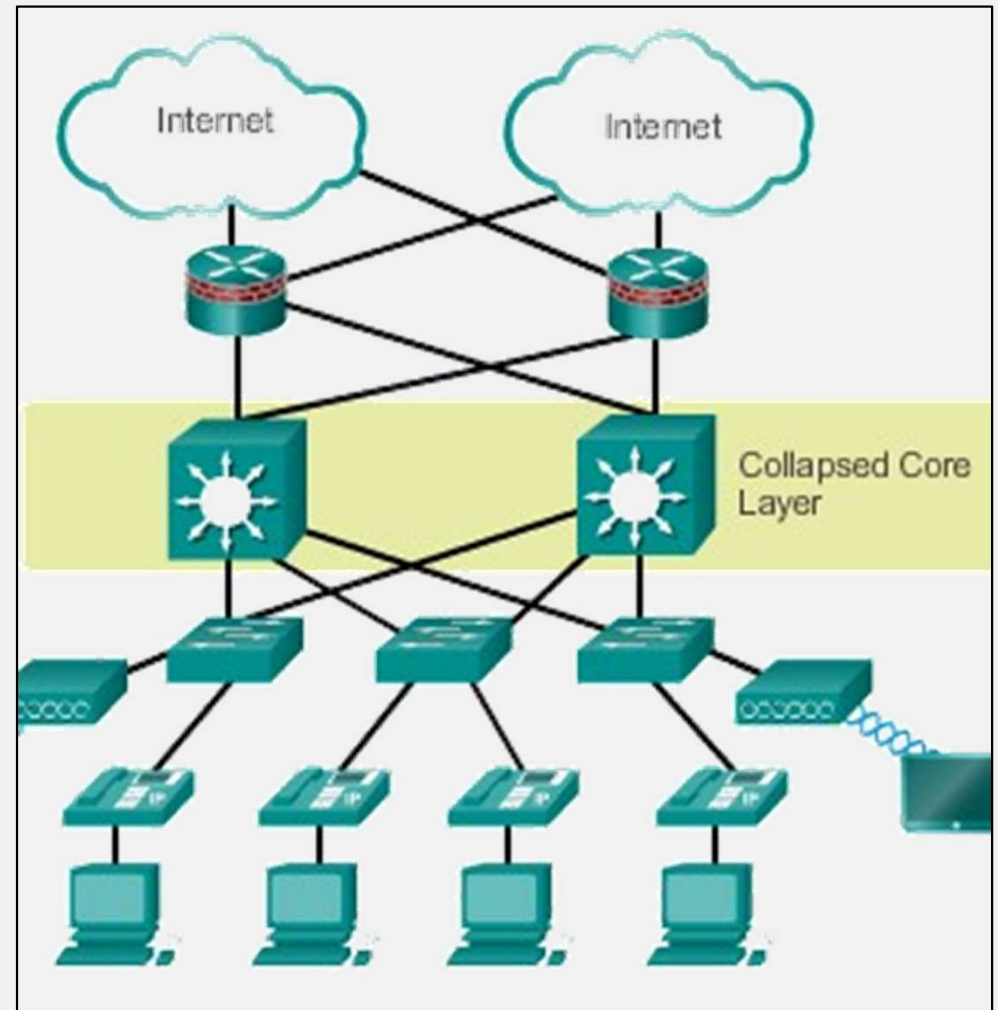
- **Distribution layer**

- Aggregation of LAN or WAN links
- Security in the form packet filtering
- Redundancy and load balancing
- Routing and Layer 3 services between LANs



Two-Tier Collapsed Core Design

- Distribution layer and core layer functions are implemented by a single device
 - Duplication is used for redundancy
- Used by smaller businesses to reduce network cost while maintaining most of the benefits of the three-tier hierarchical model

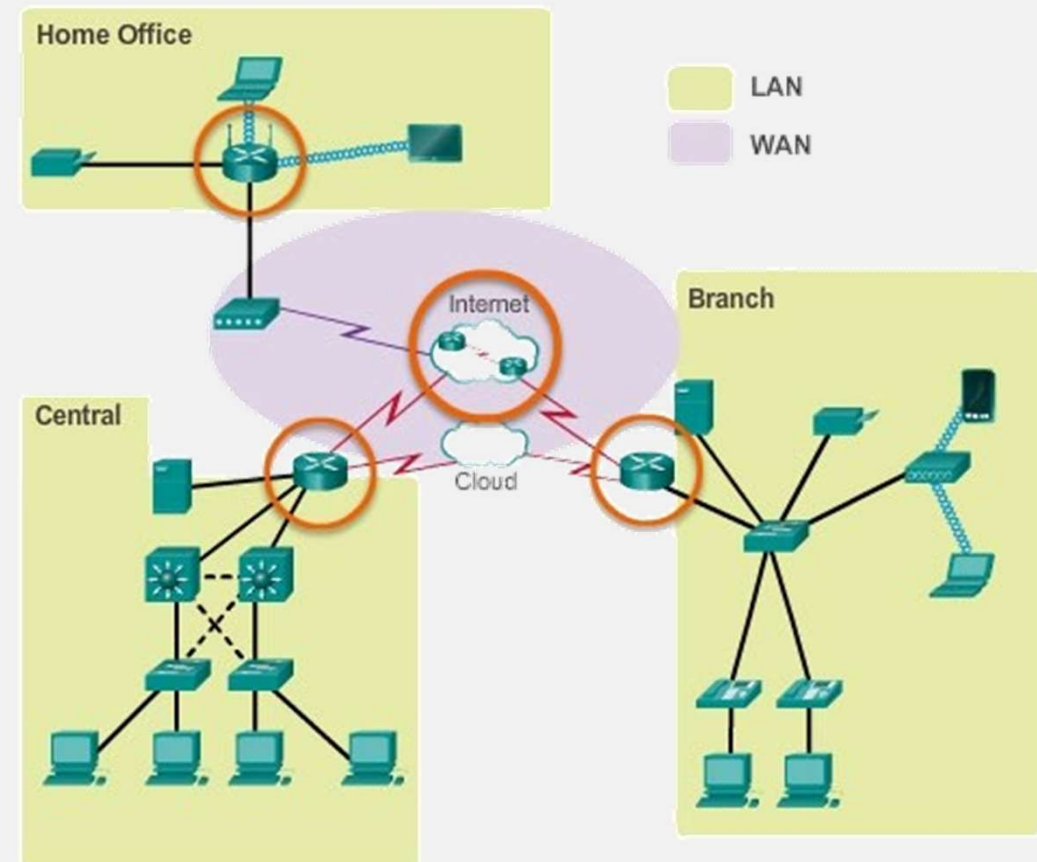


Home (or Small Business) Network

- Core, Distribution and Access layers still exist but are implemented in the same single device
 - No redundancy, only one connection
- The ISP provided Wireless Router (NOT AP / WAP) provides:
 - **Access** for user equipment (smartphone, pc, tv)
 - **Distribution** (Firewall, Routing to Internet)
 - **Core** (sharing between devices, local screencasting, media sharing)

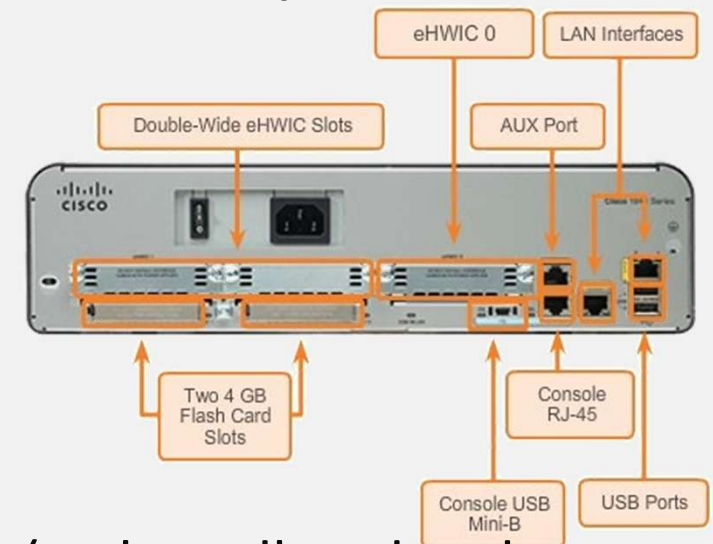
Routers Interconnect Networks

- Routers can connect multiple networks (Broadcast Domains, what we have been calling clouds)
- Routers have multiple interfaces, each connected to a different IP network (broadcast domain or cloud)



Routers are Computers

- Routers use specialised ports and network interface cards to interconnect to other networks.
- Routers are typically software based and so very slow in comparison to switches
- Routers typically have few ports
 - Low port density
- Routers have multiple port types
 - Can convert connection and protocol types (eg broadband and ethernet)



Multilayer Switching

- Also often called Layer 3 switches
- Look like layer 2 switches – need to look at part numbers sometimes
- Can act like a multiport router and a layer 2 switch
 - High port density
 - Low per port cost
 - Can build a routing table
 - Only supports a few routing protocols
 - Forwards IP packets
- Deployed in the Distribution Layer – connects LANs together

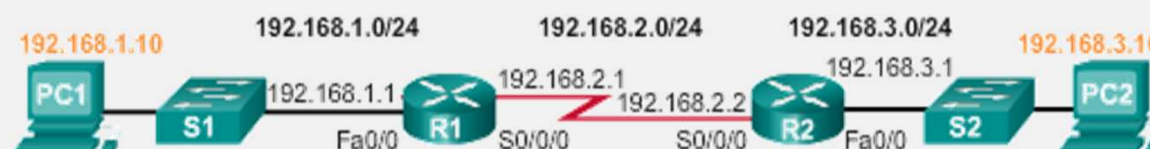


Device Addressing

- Devices are configured with IP information:
- IP address - Identifies a unique host on a local network
- Subnet mask - defines the broadcast domain (or subnet)
- Default gateway - Identifies the router that leads out of this broadcast domain (subnet)

Document Network Addressing

- Network Documentation should include at least the following in a topology diagram and addressing table:
- Device names
- Interfaces
- IP addresses and subnet mask
- Default gateways



Device	Interface	IP Address	Subnet Mask	Default Gateway
R1	Fa0/0	192.168.1.1	255.255.255.0	N/A
	S0/0/0	192.168.2.1	255.255.255.0	N/A
R2	Fa0/0	192.168.3.1	255.255.255.0	N/A
	S0/0/0	192.168.2.2	255.255.255.0	N/A
PC1	N/A	192.168.1.10	255.255.255.0	192.168.1.1
PC2	N/A	192.168.3.10	255.255.255.0	192.168.3.1

IP address allocation methods

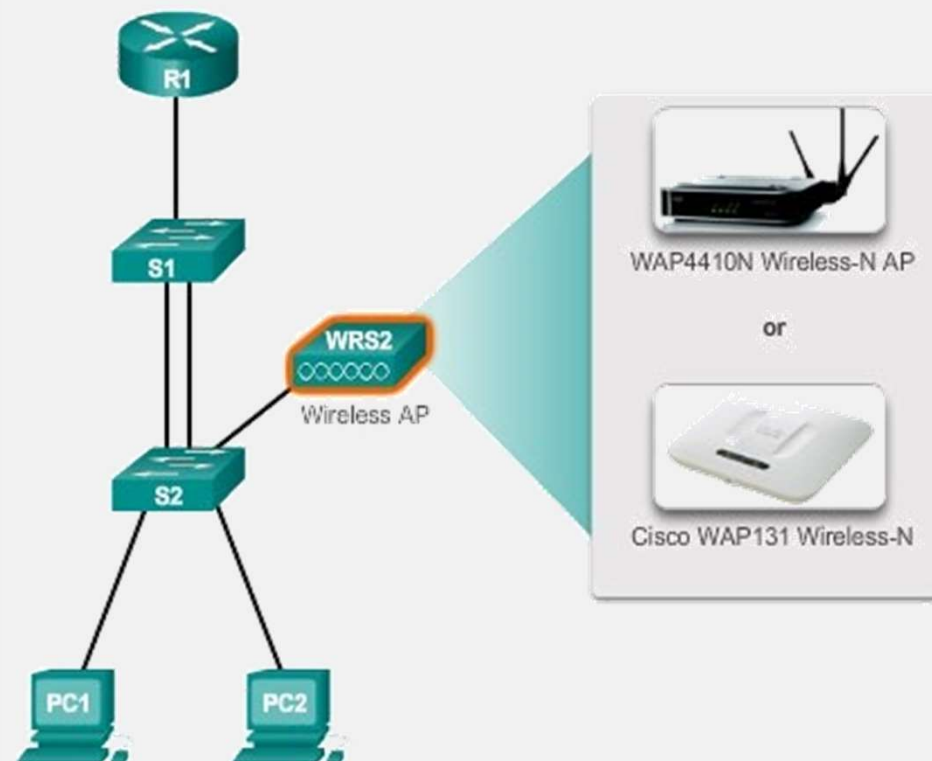
- Statically Configured IP address – The host is manually configured with an IP address, subnet mask and default gateway (probably a DNS server as well)
 - Used to identify specific network resources such as network servers and printers
 - Easy in a very small network
- Dynamically Assigned IP Address – IP Address information is dynamically allocated by a server using Dynamic Host Configuration Protocol (DHCP)
 - Most hosts acquire their IP address information through DHCP
 - DHCP services can be provided by routers or servers
 - Client addresses might change frequently
- Statically Assigned or Reserved addressing - The DHCP server can be configured to always allocate the same IP address to the same device and no other
 - All of the benefits of DHCP but no physical manual config

Designing for Scalability

- Use expandable, modular equipment or clustered devices.
- Include design modules that can be added, upgraded, and modified, without affecting the design of the other functional areas of the network.
- Create a hierarchical addressing scheme.

Expanding the Access Layer

- Access layer connectivity can be extended through wireless connectivity.
- These are AP or WAP
- Wireless routers create new networks instead of extending existing networks
 - Clue is in the title “Router”



Topologies

- Mainly covered network design characteristics with some logical design
- Will cover Physical design in the tutorial sessions

Later weeks...

- Will be covering IP addressing in detail
- Will cover network layers
- Will cover communication protocols and processes

